

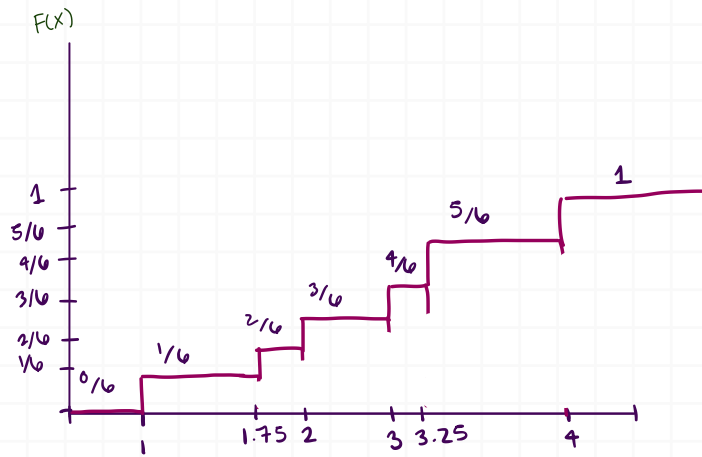
① [1, 1.75, 2, 3, 3.25, 4]

a) ECDF

$$\hat{F}(x) = \frac{\text{\# data pts} \leq x}{n} = \frac{\text{\# } n \leq x}{6}$$



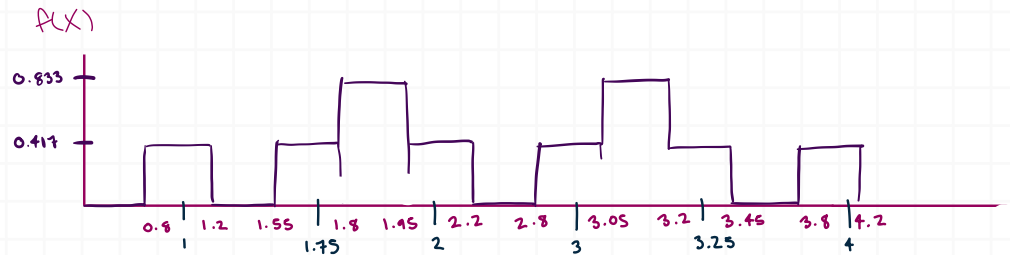
$x < 1$	$F(x) = 0/6$
$x = 1$	$1/6$
$x = 1.75$	$2/6$
$x = 2$	$3/6$
$x = 3$	$4/6$
$x = 3.25$	$5/6$
$x = 4$	$6/6$



b) KDE for $h = 0.2$

$$\hat{f}(x) = \frac{\text{\# } n \text{ w/ } h \text{ of } x}{2 \cdot h \cdot n} \rightarrow \frac{1}{2nh} = \frac{1}{2 \cdot 0.2 \cdot 6} = \frac{1}{2.4} \approx 0.417$$

x_{pt}	$(x - 0.2 \text{ to } x + 0.2)$
1	0.8 - 1.2
1.75	1.55 - 1.95
2	1.8 - 2.2
3	2.8 - 3.2
3.25	3.05 - 3.45
4	3.8 - 4.2



c) Silverman $h = 1.84 \cdot SD \cdot n^{-0.2} \rightarrow (1.84)(1.107)(0.699) \approx \boxed{1.42}$

$$\mu = \frac{1 + 1.75 + 2 + 3 + 3.25 + 4}{6} = \frac{13}{6} = 2.5$$

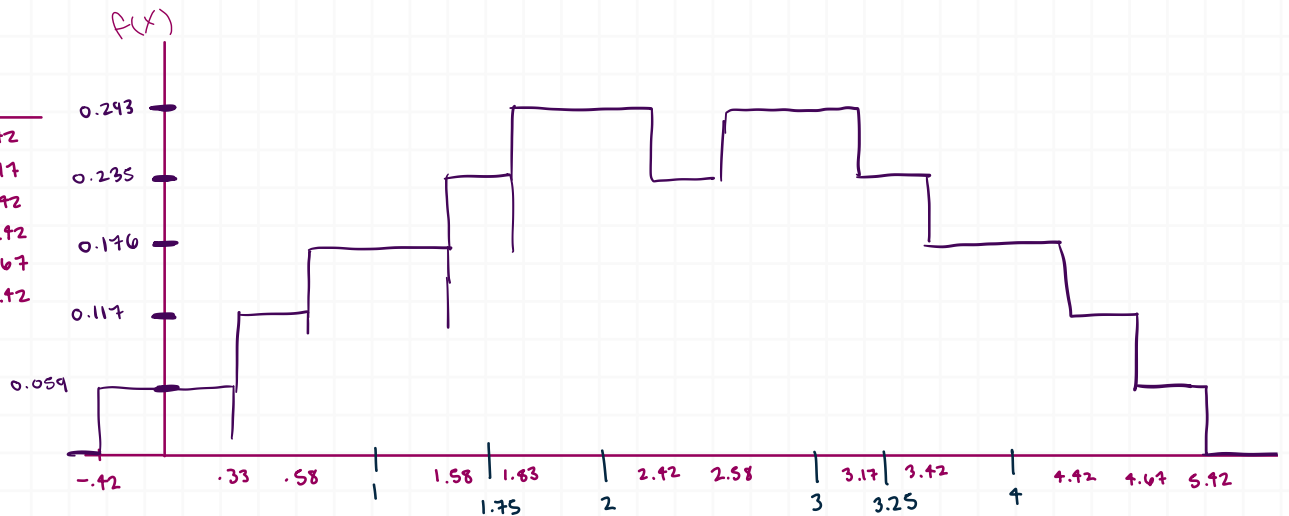
$$\sigma = \sqrt{\frac{(1-2.5)^2 + (1.75-2.5)^2 + (2-2.5)^2 + (3-2.5)^2 + (3.25-2.5)^2 + (4-2.5)^2}{6-1}} \approx 1.107$$

$$h^{-0.2} \approx 0.699$$

$$\text{height} = \frac{1}{2 \cdot 0.6 \cdot 1.42} \approx 0.059$$

d) sketch c

x_{pt}	window
1	-0.42 - 2.42
1.75	.33 - 3.17
2	.58 - 3.42
3	1.58 - 4.42
3.25	1.83 - 4.67
4	2.58 - 5.42



e) b & d comparison

When h grows from 0.2 to 1.42 each box gets wider & shorter, 0.417 drops to 0.059, but the area of each box stays $1/6$ either way. They're both still uniform kernel estimators, so both are made of flat steps & have total areas of 1. When $h = 0.2$, the boxes are narrow & barely overlap, so you get tall spikes in the data with density of 0 in the gaps. This is an example of overfitting b/c it implies certain values can never occur. With $h = 1.42$, the boxes overlap so much that they start & merge into the KDE we see above, peaking @ 2 & 3, and covering the whole range. (this could be seen as oversmoothed/underfitted since it spreads probability below 0, but we don't know what an actually accurate model looks like so won't make claim).

② $[-1, \underbrace{-0.5, 0, 0.5}_{w/h}, 1]$

★ Supporting code for Q2 is in .ipynb file in same folder as PS PDF ★

a) $h=0.6$, $x=0$; $\hat{f}(h(x))$

$$\hat{f}(0) = \frac{\text{\# pt w/ } h \text{ of } 0}{2 \cdot 5 \cdot 0.6} = \frac{3}{6} = \boxed{0.5}$$

$$b) \mathbb{E}[\hat{f}(0)] = \frac{F(0+h) - F(0-h)}{2h} = \frac{F(0.6) - F(-0.6)}{1.2} \approx \boxed{0.376}$$

$$c) f(0) = \frac{1}{\sqrt{2\pi}} \approx \boxed{0.399} \quad (\mathbb{E}=0 \text{ into gaussian formula from exp/log notebook; "the bell curve"})$$

d) No, $\mathbb{E}[\hat{f}(0)] = 0.376 < 0.399 = f(0)$. The KDE is biased, underestimating $f(0)$ by 0.023 @ 0 on average

$$e) \mathbb{E}[\hat{f}(0)] = \frac{F(0.1) - F(-0.1)}{2(0.1)} \approx \boxed{0.398}$$

gap: $0.399 - 0.398 \approx \boxed{0.001} \rightarrow$ gap shrinks towards 0 as h gets smaller ($0.6 \rightarrow 0.023$ to $0.1 \rightarrow 0.001$)

f) $\hat{f}(h)$ is only consistent as h shrinks as n grows. As $h \rightarrow 0$, the bias gets smaller and smaller (shown in part e comparison). And as n grows, the randomness goes away. Silverman's rule shrinks h w/ n slowly enough to create this consistency. With a fixed h , the bias never fully goes away, so the KDE doesn't converge to $f(x)$ no matter how large n gets.

g) The main consideration is how much you can reduce bias without having too high of a variance (tradeoff). A smaller h lowers the bias, but if it's too small relative to sample size it will overfit & will be spiky & change a lot from sample to sample. A large h has a lower variance & higher bias. So, h should shrink as n grows. It is worth it to start w/ Silverman's rule & try a couple of values of h to see which estimate is most accurate

③ $X = [5, 7, 7, 8, 10]$, $h = 1.5$

a) $X = 6, 6.5, 7, 8, 8.5$

$$\frac{1}{nh} = \frac{1}{15}$$

X	Window	(distance ≤ 1.5) pts inside	count	$\hat{f}(x)$
6	4.5 - 7.5	5, 7, 7	3	3/15
6.5	5 - 8	7, 7	2	2/15
7	5.5 - 8.5	7, 7, 8	3	3/15
8	6.5 - 9.5	7, 7, 8	3	3/15
8.5	7 - 10	8	1	1/15

b) $\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{X_i - x}{h}\right)^2\right)$

$$\hat{f}(6) = 0.151$$

$$\hat{f}(6.5) = 0.169$$

$$\hat{f}(7) = 0.178$$

$$\hat{f}(8) = 0.167$$

$$\hat{f}(8.5) = 0.151$$

★ supporting code in .ipynb file ★

c) The biggest jump in part a was from $x = 8.0$ to $x = 8.5$, where it drops from 0.2 to 0.067. Over the same stretch, the Gaussian only moves from 0.167 to 0.151. The uniform kernel is a flat box with vertical edges, so a point either contributes its full height, or nothing. From $x = 8$ to 8.5 , the two 7s go from a distance of 1 to a distance of exactly 1.5, which is right on the box edge so they both are excluded. This is what creates the drastic jump. The Gauss kernel is a bell curve with no edges, so every point contributes something and fades gradually as the distance grows. This creates a gradual decline instead of sudden jumps in this case, which gives a smoother KDE.

d) Uniform KDEs are easier to compute by hand, and they're simple/transparent, so easily explainable. It's directly related to the ECDF & each pt only affects a limited range. Gaussian is smoother, and doesn't produce artificial jumps or sudden changes that aren't real features of the data. For these reasons, the Gauss is preferred for plotting/presenting/interpreting a density at any sample size.

★ Q4 IN NOTEBOOK ★

